

Joshua Himmens

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Engineering Physics Student at The University of British Columbia

Technical Skills — **Machine Learning** | TensorFlow, Keras, PointNet, Weights and Biases (wandb), ONNX

Embedded Programming | Experienced with FreeRTOS on TMS570, RP2040, and STM32

Software Development | C, C++, Python, Java, MATLAB, Bash, Git

Particle Physics | Root, CERN grid compute, Athena

Work Experience — **Associate Machine Learning Developer** AltaML | Jan 2025 – Apr 2025 |

himmens.com/altaml

ATLAS Deep Learning Research Student TRIUMF | May 2024 – Feb 2025 | himmens.com/triumf

- Developed panoptic segmentation models for the ATLAS detector using the PointNet ML framework with Wandb, TensorFlow, Keras.
- Used CERN's grid computing to parallelize compute across thousands of nodes.
- Worked independently to develop models using cutting edge transfer learning approaches.
- Implementing models into production using NVIDIA Triton.

Command and Data Handling (CDH) Lead and Firmware Developer UBC Orbit Satellite Design Team | Oct

2023 – Present | himmens.com/orbit

- Led the CDH team to develop software to meet mission and testing objectives from ESA (European Space Agency) for the ALEASAT project.
- Managed a team of **10** firmware developers, with over **\$15 000** worth of technical infrastructure.
- Developed mission testing, function testing, and acceptance testing procedures to meet ESA and ECSS standards.
- Programmed device drivers and electrical ground support equipment (EGSE).

Embedded Firmware Developer UBC Orbit Satellite Design Team | Jan 2023 – Present | himmens.com/orbit

- Programmed device drivers, electrical ground support equipment (EGSE).
- Developed the ALEASAT Avionics Test Bench (FlatSat).
- Worked on the ALEASAT onboard computer for launch in Q1 2026.

Elections Officer Elections Alberta | Jan 2023 – Dec 2023

- Integrated as part of a team to run a safe and fair provincial election.
- Managed public interactions and vote counting.

Publications and Presentations . **Developing Machine Learning Techniques for Particle Flow in the ATLAS Experiment** at Canadian Astroparticle Physics Summer Student Talks Competition (CASST 2024) Jun 2024

- Placed 2nd of 44 presentations

JetPointnet: A Machine Learning Approach to Cell-to-Track Attribution in the ATLAS Experiment at Canadian Undergraduate Physics Conference Oct 2024

3D Particle Flow in the ATLAS Calorimeter: How to Train Your Model at TRIUMF Science Week May 2024

ALEASAT ESA "Fly Your Satellite!" Training Week Presentation at European Space Agency's ESEC-GALAXIA (Transinne Belgium) Mar 2024

Honors and Awards — **Hector J. MacLeod Award** | Apr 2024

Awarded to students who have demonstrated pioneering efforts in their chosen field of study and attained high scholastic honours.

Erich Vogt First Year Summer Research Experience (FYSRE) Award | May 2024

Awarded to promising students in physics for a 4-month research placement.

Alberta Centennial Award and Alberta Premier's Citizenship Award | May 2023

Awarded for outstanding community service.

Calgary Flames Foundation Community Involvement Scholarship | Aug 2023

Awarded for community involvement.

Julia Turnbull Leadership Award for exceptional community service | Jul 2023

Awarded for exceptional community service.

Ted Rogers Entrance Scholarship | Jun 2023

Awarded for academic achievement.

Leadership and Advocacy _____ split_job(title: role.position, company: role.organization, date: utils.strptime(role.startDate) + " - " + (if role.endDate == "present" { ui.labels.present } else { utils.strptime(role.endDate) }), actions: role.highlights, visible: true) split_job(title: role.position, company: role.organization, date: utils.strptime(role.startDate) + " - " + (if role.endDate == "present" { ui.labels.present } else { utils.strptime(role.endDate) }), actions: role.highlights, visible: true) **Additional Experience** _____ achievement(focus: "" + exp.name + "", description: exp.description, visible: true) achievement(focus: "" + exp.name + "", description: exp.description, visible: true) achievement(focus: "" + exp.name + "", description: exp.description, visible: true)